Thomas J. Stoll, Ph.D.

Auditory Neuroscience | Deep Learning | Electrophysiology

(716) 531-0903 • tom@tomstoll.com • **tomstoll.com** • linkedin.com/in/thomas-stoll • github.com/tomstoll

TECHNICAL SKILLS

Modeling & Deep Learning
- PyTorch, CNNs, Encoder-Decoder architectures, continuous biosignals
- Predictive & LME Modeling

Programming
- Python (NumPy, SciPy, MNE, Pandas, Matplotlib)
- MATLAB, Bash/Linux, Git
- High-Performance Compute (Slurm)

Biosignals & Neurotech
- EEG, ABR, ERPs, TRFs
- Biosignal DSP, Psychophysics

Hardware & Prototyping
- CAD / 3D Design, FDM/Resin 3D Printing
- Machining, Benchtop Electronics

EDUCATION

University of Rochester
Ph.D., Biomedical Engineering
Maddox Lab 2018 – 2024

University of Rochester
M.S., Biomedical Engineering
Maddox Lab 2018 – 2019

University at Buffalo
B.S., Biomedical Engineering
B.A., Mathematics
Summa Cum Laude (Both) 2014 – 2018

HONORS & AWARDS

NIH T32 Postdoctoral Traineeship
HBCS Program (T32 DC000011) 2026 – Present

ARO MidWinter Travel Award
Assoc. for Research in Otolaryngology 2020 & 2022

Dean's List
University at Buffalo 2014 – 2018

SUMMARY

Auditory neuroscientist and research engineer with over 8 years of experience specializing in human EEG recordings, signal processing, and computational modeling. Experienced in combining electrophysiology with applied PyTorch frameworks to model sensory encoding and predict brain responses to sound. Driven to translate computational neuroscience and predictive AI into clinical diagnostic systems, neural interfaces, and next-generation smart audio devices.

RESEARCH & PROFESSIONAL EXPERIENCE

Postdoctoral Research Fellow **Ann Arbor, MI**
University of Michigan (Maddox Lab) 2024 – Present

- Developed an individualizable deep neural network framework (CNN/encoder-decoder) predicting continuous human multi-channel EEG from acoustic speech stimuli (arXiv:2609.20595).
- Engineered end-to-end training and evaluation pipelines in PyTorch, optimizing loss formulations and hyperparameters across high-performance computing clusters.
- Co-invented proprietary deep learning architectures and methods for subject-specific brain response prediction (U.S. Provisional Patent 63/986,368).
- Appointed to competitive NIH T32 Institutional Postdoctoral Traineeship (T32 DC000011, HBCS Program) to investigate computational models of sensory encoding.
- Mentored new lab members, advised collaborators on EEG experimental setup and analysis, assisted in administrative tasks and regulatory compliance.

Doctoral Researcher **Rochester, NY**
University of Rochester (Maddox Lab) 2018 – 2024

- **Dissertation:** *Advancements in Stimulus and Analysis Paradigms to Study Subcortical Sound Encoding.*
- Conducted experiments investigating the place specificity of the parallel auditory brainstem response (pABR) using computational models and human electrophysiology.
- Investigated effects of selective auditory attention to speech across the subcortical and cortical auditory pathway by extending the temporal response function framework to the subcortex.
- Fabricated, adapted, and tested custom electrodes to record directly from the eardrum for high-SNR subcortical electrophysiology.
- Mentored undergraduates, assisted in regulatory compliance.

Additional Experience

Telluride Neuromorphic Cognition Engineering Workshop, **Telluride, CO** July 2025

- Competitively selected for a 3-week research workshop bringing together international leaders in neuromorphic engineering, neuroscience, and machine learning.
- Led an experimental track investigating neural signatures of selective attention switching with EEG.
- Co-led decoding pipelines reconstructing speech representations from imagined speech using intracranial EEG (iEEG).
- Assisted in deploying real-time closed-loop decoding of auditory attention via EEG to control external demonstration hardware.

Teaching Assistant — **Rochester, NY**

University of Rochester, Department of Biomedical Engineering — 2019 – 2020

- Biomedical Signals, Systems, and Imaging: Led hands-on laboratory sections, held office hours, created rubrics, and graded technical lab reports.
- Biomedical Computation and Statistics: Led computational problem sessions in MATLAB/statistics, held office hours, and evaluated coursework.

Research Assistant Intern — **Buffalo, NY**

Toshiba Stroke and Vascular Research Center (Ionita Lab) — 2015 – 2017

- Extracted 3D vascular models from CT scans and manufactured patient-specific 3D-printed phantoms for catheter-based device testing.
- Formulated testing methodologies to analyze mechanical compliance across novel 3D-printed materials through multi-material printing.

Patents

- Maddox, R. & **Stoll, T.** (2026). *Systems and Methods for Predicting Individual Brain Responses to Acoustic Stimuli Using Deep Neural Networks.* U.S. Provisional Patent Application 63/986,368, filed February 19, 2026. Patent Pending.

Publications & Preprints

- Chen, F., **Stoll, T. J.**, Jones, G. R., & Maddox, R. K. (2026). Refractoriness-Based Masking Yields Threshold-Specific Human Auditory Nerve Responses. *bioRxiv preprint*. doi:10.64898/2026.09.19.752910.
- **Stoll, T. J.**, & Maddox, R. K. (2026). A Deep Neural Network for Predicting Continuous Human EEG Across the Auditory Pathway in Response to Sound. *arXiv preprint arXiv:2609.20595 [eess.AS]*. doi:10.48550/arXiv.2609.20595 (Submitted to ICASSP 2027).
- **Stoll, T. J.**, Vandjelovic, N. D., Polonenko, M. J., Li, N. R., Lee, A. K., & Maddox, R. K. (2025). The auditory brainstem response to natural speech is not affected by selective attention. *PLOS Biology*, 23(10), e3003407. doi:10.1371/journal.pbio.3003407.
- **Stoll, T. J.**, & Maddox, R. K. (2024). Enhanced Place Specificity of the Parallel Auditory Brainstem Response: An Electrophysiological Study. *Journal of the Association for Research in Otolaryngology (JARO)*, 25, 477–489. doi:10.1007/s10162-024-00959-w.
- **Stoll, T. J.**, & Maddox, R. K. (2023). Enhanced Place Specificity of the Parallel Auditory Brainstem Response: A Modeling Study. *Trends in Hearing*, 27. doi:10.1177/23312165231205719.
- Tabaczynski, J. R., **Stoll, T.**, Shepard, L., Siddiqui, M. I., Karkhanis, N. V., Sommer, K., ... & Ionita, C. N. (2018). Use of patient specific 3D printed (3DP) neurovascular phantoms for mechanical assessment of devices used in image guided minimally invasive procedures. *SPIE Medical Imaging*, 10579, 117–125. doi:10.1117/12.2293370.

Conference Presentations

- **Stoll, T.** & Maddox, R. (2026, Dec). *A Deep Neural Network for Predicting Continuous Human EEG Across the Auditory Pathway in Response to Sound*. NeurIPS 2026 Workshop on Foundation Models for the Brain and Body (BrainBodyFM), Sydney, Australia.
- **Stoll, T.** & Maddox, R. (2026, Feb). *The Effect of Electrode Placement on the Auditory Brainstem Response*. Poster, Association for Research in Otolaryngology (ARO), San Juan, PR.
- **Stoll, T.**, Casale, J., & Maddox, R. (2025, Feb). *Towards a Generalized Deep Neural Network Model of Human EEG Responses to Sounds*. Poster, ARO, Orlando, FL.
- **Stoll, T.** & Maddox, R. (2024, Feb). *Exploring a Possible EEG Marker of Anticipatory Processing in a Natural Speech-Based Selective Attention Task*. Poster, ARO, Anaheim, CA.
- **Stoll, T.** & Maddox, R. (2023, Feb). *Investigating the Effect of Selective Attention on Speech Encoding from Auditory Nerve to Cortex*. Poster, ARO, Orlando, FL.
- **Stoll, T.** & Maddox, R. (2022, Jul). *Place Specificity of the Parallel Auditory Brainstem Response*. Poster, Gordon Research Conference: Auditory System, Smithfield, RI.
- **Stoll, T.** & Maddox, R. (2020, Jan). *Modeling Improved Place Specificity in the Parallel Auditory Brainstem Response*. Poster, ARO, San Jose, CA.